Decision Making Under Uncertainty

ELG 5218 – From Bayesian Decision Theory to Markov Decision Processes

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Outline: Chapter 34 – Decision Making Under Uncertainty

1. **Statistical Decision Theory: From Probabilities to Actions**
   - Agents, actions, loss functions, risk
   - Frequentist vs. Bayesian decision theory
2. **One-Shot Decision Examples**
   - Classification (0-1 loss $\rightarrow$ MAP), Regression (ℓ_2 loss $\rightarrow$ MMSE)
3. **Influence Diagrams & Value of Information**
   - Oil wildcatter example, decision trees
4. **Sequential Decisions: Contextual Bandits**
   - Exploration–exploitation, UCB, Thompson sampling
5. **Markov Decision Processes (MDPs)**
   - Value functions, Bellman equations, POMDP
6. **Planning: Value Iteration & Policy Iteration**

Part 1: Statistical Decision Theory

From beliefs to actions

Review: What is Bayesian Inference?

Example: Medical Diagnosis

- **Prior:** A doctor initially estimates a 1% probability that a patient has a certain disease.
- **New Evidence:** The patient undergoes a series of medical tests (e.g., a CT scan).
- **Posterior:** Based on the test results, the doctor updates the probability of the disease to 60%.

Conclusion

The Bayesian approach is essentially **"updating prior beliefs based on new evidence."**

Key Question

If you are the doctor,
Should you operate on the patient?

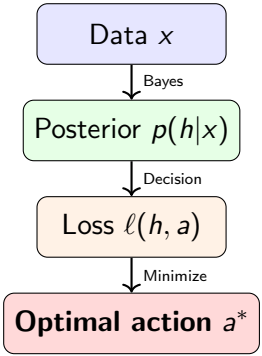
From Inference to Action

The Problem:

- Bayesian inference gives us the posterior
- But a posterior is **not a decision**
- We need to choose an **action** $a \in \mathcal{A}$
- Decision theory = turning beliefs into **optimal actions**

Key components:

- **Agent**: the decision maker
- **State of nature**: $h \in \mathcal{H}$ (truth is unknown)
- **Action**: $a \in \mathcal{A}$ (what we choose)
- **Loss function**: $\ell(h, a)$ (cost of action a when truth is h - **The Price of Failure**)



Policy, Risk, and the Optimal Decision

Policy (decision rule): A mapping from observations to actions

$$\delta : \mathcal{X} \to \mathcal{A}, \qquad a = \delta(x)$$

Risk: The expected loss under a policy

$$R(\delta) = \mathbb{E}\left[\ell(h, \delta(X))\right]$$

Optimal policy: Minimizes the risk

$$\delta^*(\cdot) = \underset{\delta}{\operatorname{argmin}} \; R(\delta)$$

Key Question

How do we define the expectation $\mathbb{E}[\cdot]$? This leads to two approaches: **Frequentist**(constant) vs. **Bayesian**(random variable).

Two Views of Risk

Frequentist Risk

Treat h as fixed, X as random:

$$r(\delta|h) = \mathbb{E}_{p(x|h)}[\ell(h, \delta(x))]$$

Problem: h is unknown, so we cannot compute this.

Solutions:

- Minimax: $\min_\delta \max_h r(\delta|h)$ (pessimistic – assumes adversary)
- Bayes risk: put prior $\pi(h)$

Bayesian Posterior Risk

Treat data x as observed, h as random:

$$\rho_\pi(a|x) = \mathbb{E}_{p(h|x)}[\ell(h, a)]$$

Optimal action (current estimation):

$$\delta^*(x) = \underset{a \in \mathcal{A}}{\operatorname{argmin}} \ \rho_\pi(a|x)$$

Equivalently, **maximum expected utility**:

$$\delta^*(x) = \underset{a \in \mathcal{A}}{\operatorname{argmax}} \ \mathbb{E}_h[U(h, a)]$$

Two Views of Risk

The Core Difference

Frequentist: Evaluate a **fixed rule** across *all possible* data.
Bayesian: The data is here; make the **best decision** *right now*.

Frequentist

The Administrator (Pre-data)

- **Focus:** A fixed policy $\delta(X)$.
- **Example:** "If CT shadow > 2cm, always operate."
- **Risk:** Averages over **all possible future** CT scans.

Bayesian

The ER Doctor (Post-data)

- **Focus:** A specific action a.
- **Example:** "Given **this exact** CT scan, do I operate?"
- **Risk:** Expected loss for **this specific** patient.

Key Result: Bayesian Approach is Frequentist-Optimal

Bayesian path E[h] then E[X]

$$\begin{tikzcd}
 & \boxed{\rho(\delta|x) = \mathbb{E}_{h\sim p(h|x)}[\ell]} \arrow[dr, "\text{avg over } x"] & \\
\boxed{\ell(h, \delta(x))} \arrow[ur, "\text{avg over } h|x"] \arrow[dr, "\text{avg over } x|h"'] & & \boxed{R_B(\delta) = \mathbb{E}_{(x,h)}[\ell]} \\
 & \boxed{r(\delta|h) = \mathbb{E}_{x\sim p(x|h)}[\ell]} \arrow[ur, "\text{avg over } h"'] &
\end{tikzcd}$$

Frequentist path E[X] then E[h]

Key Result: Both Paths Lead to the Same Risk

By **Fubini's theorem** (exchanging order of integration):

$$R_B(\delta) = \mathbb{E}_{p(x)}[\rho(\delta|x)] = \mathbb{E}_{p(h)}[r(\delta|h)]$$

Conclusion

Since both paths yield the same global risk, **nothing is lost**. But practically:

- **Frequentist:** Only tells you how to *evaluate* a rule.
- **Bayesian:** Actually tells you *how to construct it based on your observation*.

Part 2: One-Shot Decision Examples

Classification and Regression as Decision Problems

Classification = Decision with 0-1 Loss

Setup: $\mathcal{H} = \mathcal{Y} = \{1, \ldots, C\}$, $\mathcal{A} = \mathcal{Y}$

Zero-one loss:

$$\ell_{01}(y^*, \hat{y}) = \mathbb{I}(y^* \neq \hat{y}) = \begin{cases} 0 & \text{if } \hat{y} = y^* \\ 1 & \text{if } \hat{y} \neq y^* \end{cases}$$

Posterior expected loss:

$$\rho(\hat{y}|x) = p(\hat{y} \neq y^*|x) = 1 - p(y^* = \hat{y}|x)$$

Optimal action (minimize the loss):

$$\boxed{\delta(x) = \underset{y \in \mathcal{Y}}{\operatorname{argmax}}\ p(y|x) \quad \text{(Maximum A Posteriori estimate)}}$$

Takeaway: The familiar "pick the most probable class" rule is the **Bayes optimal** decision under 0-1 loss.

Regression = Decision with ℓ_2 Loss

Setup: $h \in \mathbb{R}$, action $y \in \mathbb{R}$

Quadratic loss: $\ell_2(h, y) = (h - y)^2$
Posterior expected loss:

$$\rho(y|x) = \mathbb{E}\left[(h-y)^2 \,\middle|\, x\right] = \mathbb{E}[h^2|x] - 2y\,\mathbb{E}[h|x] + y^2$$

Optimal action (set derivative to zero - find the center of gravity):

$$\frac{\partial}{\partial y}\rho(y|x) = -2\mathbb{E}[h|x] + 2y = 0$$

$$\boxed{\delta(x) = \mathbb{E}[h|x] = \int h\, p(h|x)\, dh \quad \text{(Posterior Mean / MMSE)}}$$

A Summary Table

Loss Function	Optimal Estimate	Name
0-1 loss $\mathbb{I}(h \neq a)$	Posterior mode	MAP
ℓ_2 loss $(h-a)^2$	Posterior mean	MMSE
ℓ_1 loss $\lvert h-a \rvert$	Posterior median	–

Engineering Tip

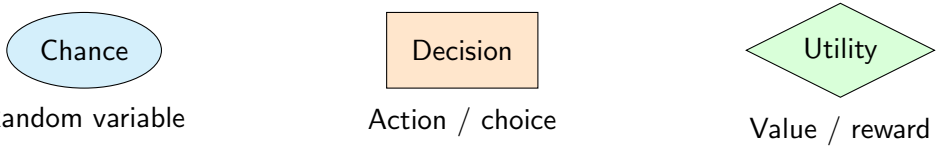
The loss function is **your design choice**. There is no single "correct" answer—you choose the loss based on which errors are most unacceptable in your specific application.

Part 3: Influence Diagrams

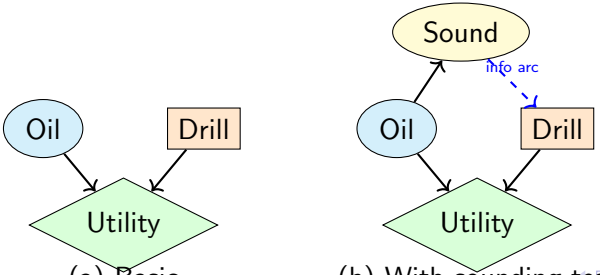
Decision under uncertainty with information

Influence Diagrams: Graphical Notation

Extend Bayesian networks with two new node types:



Oil Wildcatter Problem (Raiffa 1968):



(a) Basic (b) With sounding test

Oil Wildcatter: Problem Setup

Decision: Drill ($d = 1$) or don't drill ($d = 0$)?
States of nature: $o \in \{\text{dry, wet, soaking}\}$

Prior beliefs: $p(o) = [0.5,\ 0.3,\ 0.2]$

Utility table $U(d, o)$:

	$o = \text{dry}$	$o = \text{wet}$	$o = \text{soaking}$
$d = 0$ (don't drill)	0	0	0
$d = 1$ (drill)	-70	$+50$	$+200$

Expected utility with no information:

$$EU(d = 0) = 0$$
$$EU(d = 1) = 0.5 \times (-70) + 0.3 \times 50 + 0.2 \times 200 = \mathbf{20}$$

$\Rightarrow$ **Optimal: drill!** $\quad \text{MEU} = \max\{0, 20\} = 20$

Oil Wildcatter: Adding a Noisy Test

Sounding test: noisy measurement $s \in \{0, 1, 2\}$ about the oil state.
Sensor model $p(s|o)$:

	$s = 0$ (diffuse)	$s = 1$ (open)	$s = 2$ (closed)
$o =$ dry	0.6	0.3	0.1
$o =$ wet	0.3	0.4	0.3
$o =$ soaking	0.1	0.4	0.5

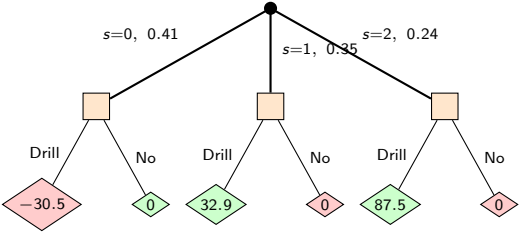
After Bayesian update The test saves you from drilling a dry hole.:

Observation	$EU(d{=}0\|s)$	$EU(d{=}1\|s)$	Optimal
$s = 0$ (diffuse)	0	-30.5	Don't drill
$s = 1$ (open)	0	$+32.9$	**Drill**
$s = 2$ (closed)	0	$+87.5$	**Drill**

Decision Tree & Value of Perfect Information

Prior marginal on test outcome:

$$p(s) = \sum_o p(o)\, p(s|o) = [0.41,\ 0.35,\ 0.24]$$



MEU with test:

$$\begin{aligned} \text{MEU} &= 0.41 \times 0 + 0.35 \times 32.9 \\ &\quad + 0.24 \times 87.5 \\ &= \mathbf{32.2} \end{aligned}$$

Value of Perfect Information (VPI)

$$\text{VPI} = \text{MEU}_{\text{with test}} - \text{MEU}_{\text{without}} = 32.2 - 20 = \ 12.2$$

Do the test if it costs less than $122k$.

Part 4: Sequential Decisions

From one-shot to multi-armed bandits

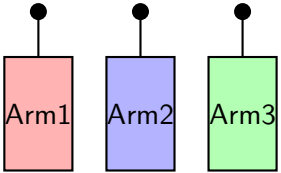
From One-Shot to Sequential Decisions

One-shot (e.g., A/B testing):

- Try both options n_0, n_1 times
- Pick the winner, deploy forever
- No feedback during testing

Sequential (bandits):

- At each step t: choose action a_t
- Observe reward r_t
- **Update policy** based on feedback
- Key: "earning while learning"



Multi-armed bandit
Which arm pays the most?

The Exploration–Exploitation Tradeoff

The fundamental dilemma:

- **Exploit:** Pick the action you *think* is best $\rightarrow$ high short-term reward
- **Explore:** Try uncertain actions $\rightarrow$ learn, potentially find better ones
- Too much exploitation $\rightarrow$ miss the true best action
- Too much exploration $\rightarrow$ waste time on suboptimal actions

Contextual bandit: adds context/state s_t (e.g., user features)

$$a_t \sim \pi_t(a_t|s_t), \quad r_t \sim p_R(r_t|s_t, a_t), \quad R(s,a) = \mathbb{E}[r|s,a]$$

Applications:

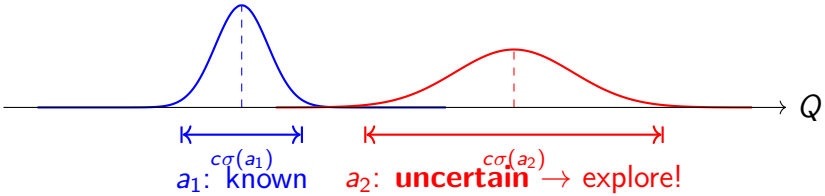
- Online advertising: state = user features, action = which ad to show
- Clinical trials: state = patient features, action = treatment
- Recommendation systems: state = user history, action = content

Solution 1: Upper Confidence Bound (UCB)

Principle: "Optimism in the face of uncertainty" - try a new restaurant!
Maintain an **optimistic estimate** of each action's reward:

$$\tilde{R}_t(a) = \underbrace{\hat{\mu}_t(a)}_{\text{estimated mean}} + \underbrace{c\,\hat{\sigma}_t(a)}_{\text{exploration bonus}}$$

Choose greedily: $a_t = \text{argmax}_a \tilde{R}_t(a)$



Actions with high uncertainty get a large bonus → encourages exploration.
As we collect data, $\hat{\sigma}$ shrinks → gradually shifts to exploitation.

Solution 2: Thompson Sampling

Idea: Sample parameters from the posterior, then act greedily.

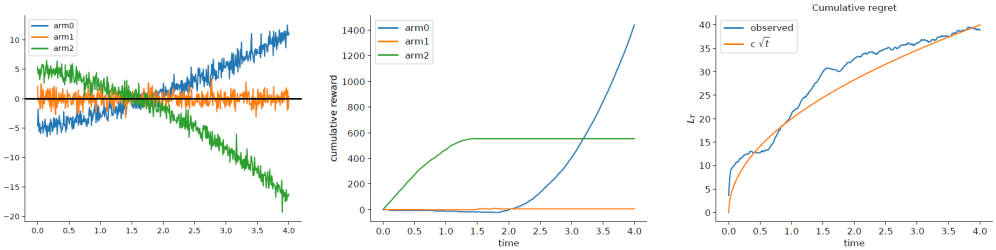
Algorithm:

1. Sample $\tilde{\theta}_t \sim p(\theta|h_t)$ (Draw a sample from the belief of each arm)
2. Pick $a_t = \text{argmax}_{a'} R(s_t, a'; \tilde{\theta}_t)$ (Pull the arm with the highest sampled value)

Why this works:

- If posterior is **uncertain** $\rightarrow$ samples are diverse $\rightarrow$ automatic exploration
- As uncertainty decreases $\rightarrow$ samples concentrate $\rightarrow$ exploitation
- Extremely simple to implement

Thompson Sampling: Empirical Performance



(a) (b) (c)

Figure 34.10: Illustration of Thompson sampling applied to a linear-Gaussian contextual bandit. The context has the form $\boldsymbol{s}_t = (1, t, t^2)$. (a) True reward for each arm vs time. (b) Cumulative reward per arm vs time. (c) Cumulative regret vs time. Generated by thompson_sampling_linear_gaussian.ipynb.

Thompson sampling on a linear-Gaussian contextual bandit: (a) true reward per arm, (b) cumulative reward, (c) cumulative regret grows more slowly, confirming more efficient learning and reduced mistakes.

Part 5: Markov Decision Processes

When actions change the state of the world

From Bandits to MDPs: What Changes?

	Contextual Bandit	**MDP**
States evolve	Independently of actions	**Depend on actions**
Transition	$s_t \sim p(s_t \mid s_{1:t-1})$	$s_{t+1} \sim p(s' \mid s_t, a_t)$
Key challenge	Explore vs. exploit	+ Long-term planning

Critical difference: In an MDP, your action today affects which states you visit tomorrow.

- A robot moving left reaches different rooms than moving right
- A drone pursuing a target changes relative geometry (part 3)
- Must consider **long-term consequences**, not just immediate reward

MDP: Formal Definition

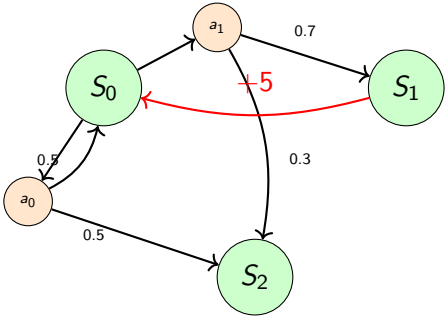
A Markov Decision Process is a tuple $\langle \mathcal{S}, \mathcal{A}, p, R, p_0 \rangle$:

- $\mathcal{S}$: set of **states** (discrete or continuous)
- $\mathcal{A}$: set of **actions**
- $p(s'|s, a)$: **transition model** (probability of next state given current state and action)
- $R(s, a)$: **reward function** (expected immediate reward)
- p_0: **initial state distribution**

Policy: $\pi(a|s)$ – probability of choosing action a in state s
Trajectory: $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, \ldots)$

Markov property: future depends only on *current* state, not history.

MDP Example: Finite State Machine



Numbers on edges = transition probabilities
Red wiggly arrow = non-zero reward

States (green circles), actions (orange circles). The agent chooses an action in each state; the environment responds with a next state (stochastically) and a reward.

Returns and Discount Factor Gamma γ

Return (reward-to-go from time t):

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}$$

Recursive form: $G_t = r_t + \gamma G_{t+1}$

Discount factor (Inflation) $\gamma \in [0, 1]$:

$\gamma = 0$	Completely **myopic**: only cares about immediate reward
$\gamma = 1$	Infinitely **far-sighted**: all future rewards weighted equally
$\gamma = 0.9$	**Good balance**: prefers near-term but still plans ahead

Value Functions: V, Q, and A

State-value function (expected return from state s under policy π, don't know the first step):

$$V^\pi(s) = \mathbb{E}_\pi[G_0 \mid s_0 = s] = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, s_0 = s\right]$$

Action-value function (Q-function, know the first action):

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_0 \mid s_0 = s, a_0 = a]$$

Advantage function:

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

Relationships:

- $V^\pi(s) = \mathbb{E}_{\pi(a|s)}[Q^\pi(s, a)]$ (V is the policy-weighted average of Q)
- $A^\pi(s, a)$ can be positive or negative; $\mathbb{E}_{\pi(a|s)}[A^\pi(s, a)] = 0$
- **Optimal policy:** Always select the action with the max Q-value in every state.

Bellman Optimality Equations

The optimal value functions V^*, Q^* satisfy:

Bellman Optimality for V^* (Recursion)

$$V^*(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} p(s'|s,a)\, V^*(s') \right]$$

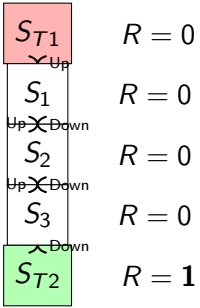
Bellman Optimality for Q^*

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} p(s'|s,a) \max_{a'} Q^*(s',a')$$

Meaning: Optimal value = immediate reward + discounted best future value.
Optimal policy (derived from Q^*):

$$\pi^*(s) = \operatorname*{argmax}_a Q^*(s,a) = \operatorname*{argmax}_a \left[R(s,a) + \gamma \mathbb{E}_{p(s'|s,a)}[V^*(s')] \right]$$

Example: 1D Grid World



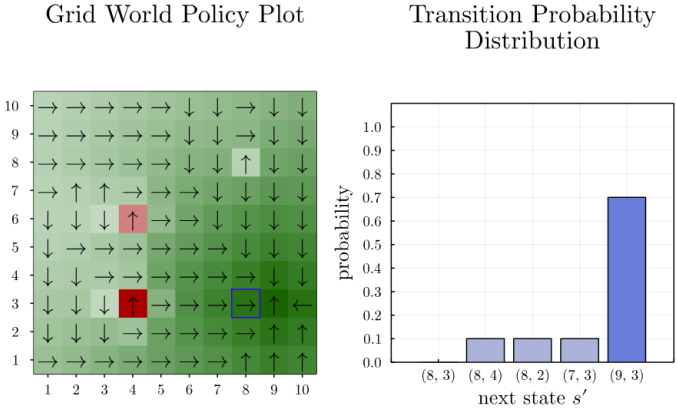
Goal: reach S_{T2} (reward $= 1$). All other rewards $= 0$. S_{T1}, S_{T2} are absorbing (terminal) states.

Optimal Q-values $Q^*(s, a)$ for different γ:

	$\gamma = 0$		$\gamma = 1$		$\gamma = 0.9$	
	Up	Down	Up	Down	Up	Down
S_1	0	0	1.0	1.0	0	0.81
S_2	0	0	1.0	1.0	0.73	0.9
S_3	0	1.0	1.0	1.0	0.81	1.0

$\gamma = 0$: only sees immediate reward (myopic)
$\gamma = 1$: no guidance – all paths eventually reach goal
$\gamma = 0.9$: **prefers shortest path** (discounting encourages speed)

Grid World: Optimal Policy and Transition Model



Left: value iteration policy on a 10×10 grid (green = high value, red = penalty). Right: stochastic transition distribution from state $(8, 3)$—the agent moves to the intended cell with $p = 0.7$.

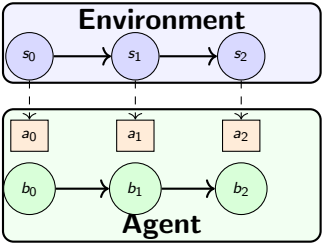
Partially Observable MDP (POMDP)

Key difference from MDP:

- Agent does **not** observe true state s_t
- Instead receives noisy observation $o_t \sim p(o|s_t, a_t)$
- Must maintain a **belief state**: $b_t = p(s_t|h_t)$
- $h_t = (a_1, o_1, \ldots, a_{t-1}, o_t)$ is the history

Solution: Convert to a **belief-state MDP**

- State = belief distribution b_t
- Transition = Bayesian update
- Computationally intractable in general



E.g., UAV tracking: true target position s_t is unknown; radar gives noisy obs o_t.

Part 6: Planning in MDPs

When the model is known

Value Iteration (VI)

Idea: Iteratively apply Bellman backup until convergence.

Value Iteration Update

$$V_{k+1}(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} p(s'|s,a)\, V_k(s') \right]$$

Convergence guarantee (Find treasure box in Gatineau Park):

$$\max_s |V_{k+1}(s) - V^*(s)| \leq \gamma \cdot \max_s |V_k(s) - V^*(s)|$$

Error shrinks by factor γ each iteration $\Rightarrow$ **guaranteed convergence** to true optimal.

After convergence: extract optimal policy using

$$\pi^*(s) = \operatorname*{argmax}_a \left[R(s,a) + \gamma \sum_{s'} p(s'|s,a)\, V^*(s') \right]$$

Policy Iteration (PI)

Idea: Alternate between evaluation and improvement. (**Commuting to uOttawa**)

Step 1 – Policy Evaluation: Given policy π, solve for V^π:

$$\boldsymbol{v} = \boldsymbol{r} + \gamma \boldsymbol{T}\boldsymbol{v} \quad \Rightarrow \quad \boldsymbol{v} = (\boldsymbol{I} - \gamma \boldsymbol{T})^{-1}\boldsymbol{r}$$

where $T(s'|s) = \sum_a \pi(a|s)\, p(s'|s,a)$ and $r(s) = \sum_a \pi(a|s)\, R(s,a)$.

Step 2 – Policy Improvement: Make policy greedy w.r.t. V^π:

$$\pi'(s) = \operatorname*{argmax}_a \left[R(s,a) + \gamma \mathbb{E}[V^\pi(s')]\right]$$

Policy improvement theorem: $V^{\pi'}(s) \geq V^\pi(s)$ for all s.

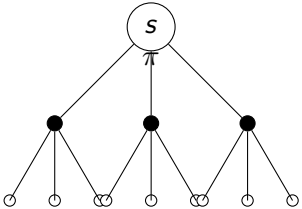
Iteration:

$$\pi_0 \xrightarrow{E} V^{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} V^{\pi_1} \xrightarrow{I} \cdots \xrightarrow{I} \pi^* \xrightarrow{E} V^*$$

Converges in finite iterations (at most $|\mathcal{A}|^{|\mathcal{S}|}$ deterministic policies).

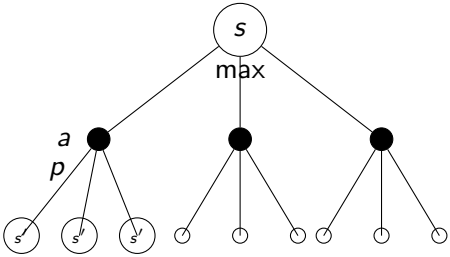
Backup Diagrams: Policy Evaluation vs. Value Iteration

Policy Evaluation



$\mathbb{E}_\pi[\cdot]$ (average)

Value Iteration



$\max_a$ (best action)

- **PE: averages** over all actions according to $\pi \rightarrow$ computes V^π
- **VI (greedy):** takes the **max** over all actions $\rightarrow$ computes V^* (ignore the action probabilities)
- Open circles = states, filled circles = (state, action) pairs

Chapter 34: Summary

1. **Decision theory** turns jugement (posterior beliefs) into optimal actions via loss minimization
2. **Different losses yield $\rightarrow$ different optimal actions:**
   0-1 loss $\rightarrow$ MAP, ℓ_2 loss $\rightarrow$ posterior mean
3. **Value of information:** quantifies whether gathering more data is worth the cost (Oil Wildcatter: VPI = \$122k)
4. **MDPs:** actions change the future $\rightarrow$ need long-term planning
   Key: value functions V, Q, A and Bellman equations
5. **Planning:** Value Iteration (Bellman backup), Policy Iteration (evaluate + improve)

Next: Chapter 35 – Reinforcement Learning (model is **unknown**)